Higher Colleges of Technology

كليات التقنية العليا

CDS 2413- PROGRAMMING FOR DATA SCIENCE

Week 13,14

Week 13,14	CLO 4: Apply data version control tools in a team environment while solving real-world business problems

Document Revision CONTROL (DRC)

Version	Author	Effective Date	Change Description	DRC No
1.0	Mohammed Abdul Rahim	13/11/2023	Define the first version	001

X.Y
X: Major change
Y: minor change

Lecture Outline

- Version Control and its importance
- Popular Version Control
- Repositories
- Files States in Git
- Cloning the Repositories
- Push and Pull
- Branching

LECTURE OBJECTIVES

- At the end of the lecture, the student should be able to do the following:
  - Understand the objective of version control, its importance, and popular tools.
  - Understand the different types of repositories.
  - Capable of examining the different file states in Git
  - Investigate the purpose of cloning repositories between local and remote repositories.
  - Investigate the methodology to push and pull the data.
  - Discover the various branching methods used in version control.

Version Control

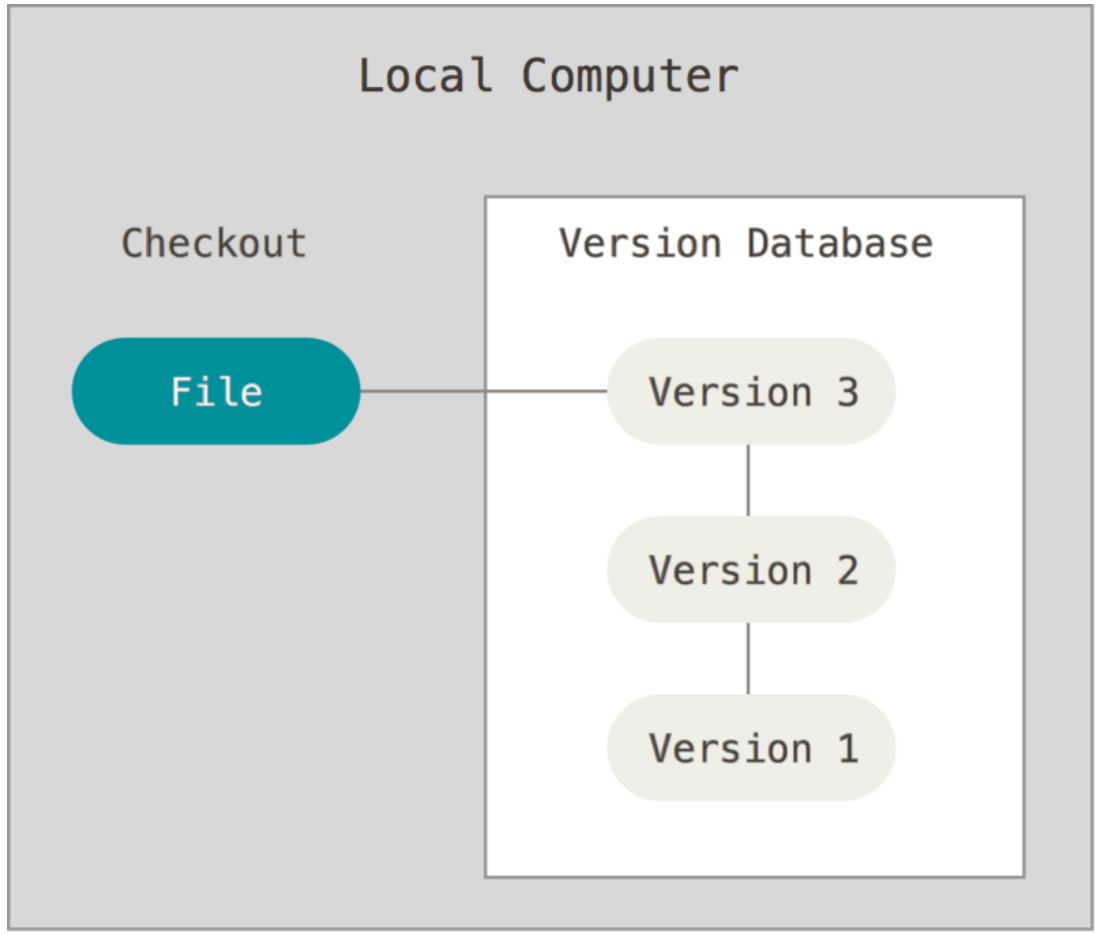
Introduction

- How do you prevent multiple team members overwriting each other's code in a team programming environment?

Introduction

Versioning Control(التحكم في الإصدار)

- Version control is the process of tracking changes to a file over time.
- it allows teams to
  - collaborate effectively,
  - experiment with new ideas, and
  - roll back changes if necessary.
- For example, in a software development team, version control enables multiple developers to work on the same project simultaneously without overwriting each other's work.



Importance of Versioning Control

- **Collaboration(تعاون)**
  - Allows multiple developers to work on the same project simultaneously,
- **Experimentation(التجريب)**
  - Makes it easy to experiment with new ideas and features
- **Reversibility(تفريغ)**
  - Allows developers to roll back changes and revert to a previous version of the code.
- **Auditability(مراجعة)**
  - Provides a complete history of all changes made to the code, which can be useful for auditing and debugging purposes.

Role of Version Control in Software Development

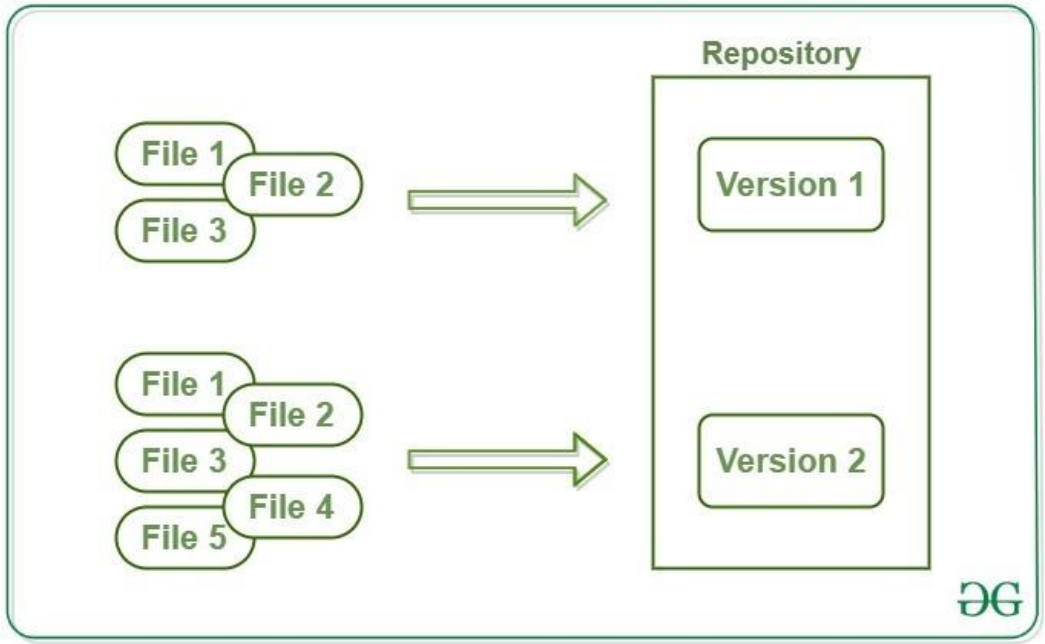
- **Manage the development process**
  - Used to track the progress of development,
  - Identify and resolve conflicts, and
  - Ensure that all changes are properly reviewed and tested.
- **Improve code quality**
  - Helps developers to review and test code thoroughly, and
  - To identify and fix bugs early on.
- **Release software faster and more reliably**
  - Makes it easier to release software frequently and
  - Reliably, by providing a way to track changes and roll back if necessary.

Popular Version Control Systems

- **Git**
  - A distributed version(النسخة الموزعة) control system that is widely used in open source and commercial software development.
- **Subversion**
  - A centralized version(نسخة مركزية) control system that is popular among enterprise users.
- **Mercurial**
  - A distributed version control system that is similar to Git, but with a few key differences.

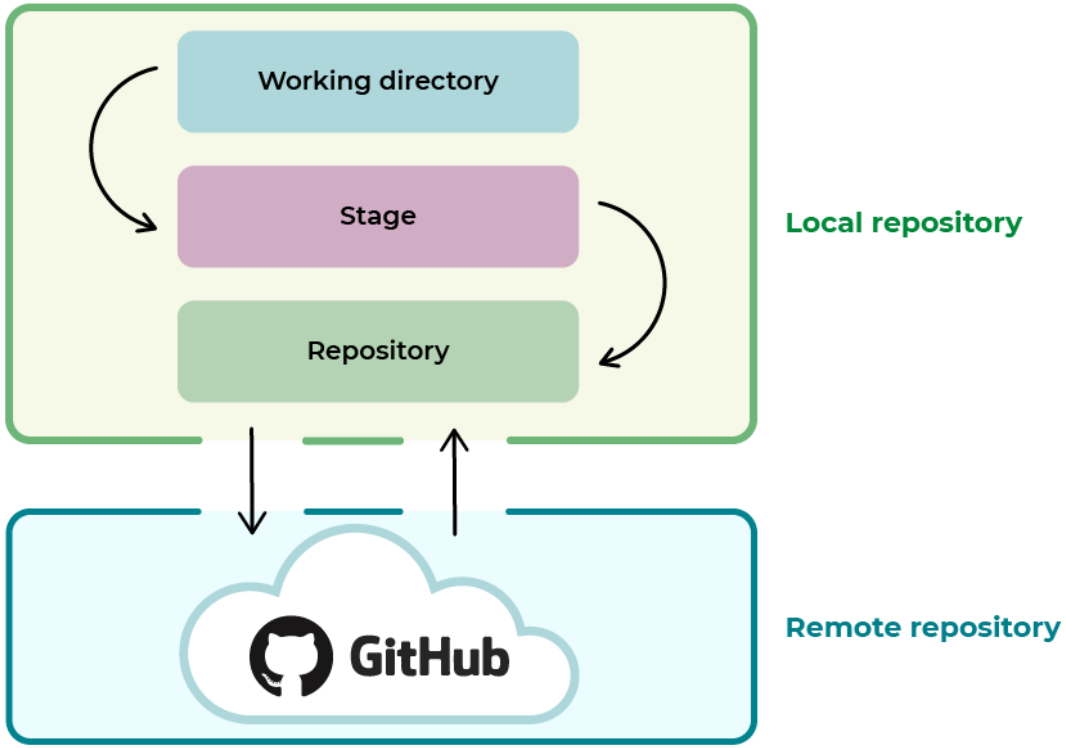
Repositories(مخزن)

- A Git repository is a collection of files and folders that are tracked by Git.
- It can be stored locally on your computer, or remotely on a server.
- To create a new Git repository, you can use the following command:
  - **git init**



Repositories(مستودع)

- Types of repository
- Local Repository – Git
  - **stored on your local computer.**
  - Used for **software development** work
- Remote Repository – Git Hub
  - **stored on a server**
  - used for **collaboration and deployment**

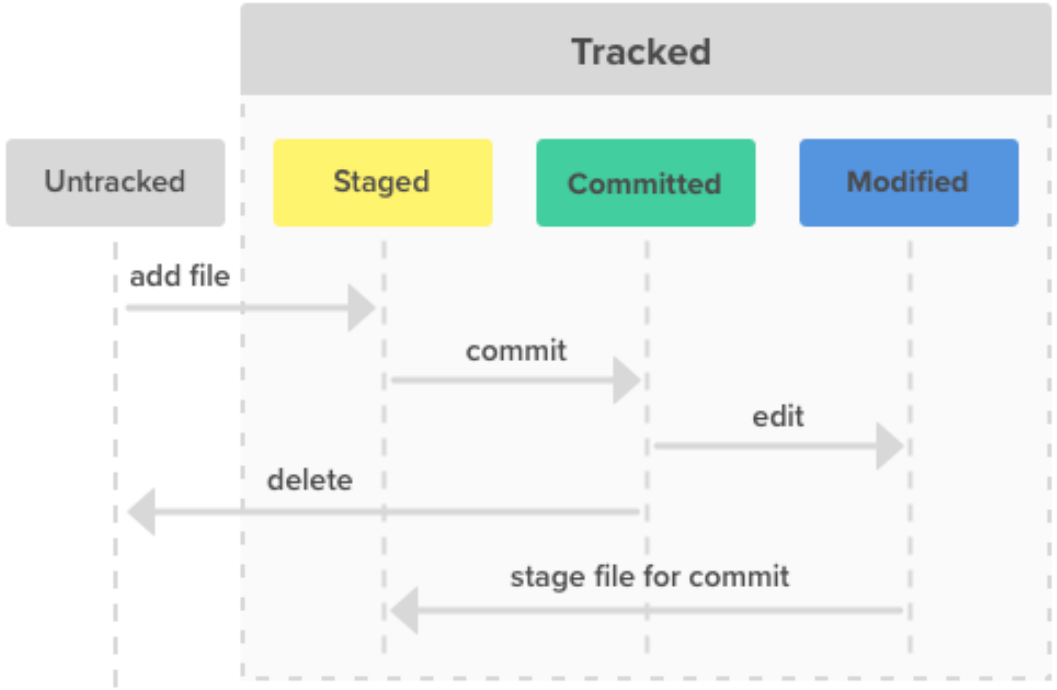


Local vs Remote

Feature	Git	GitHub
Type	Version control system	Hosting service for Git repositories
Installation	Requires installation on your local computer	Accessed through a web browser
Storage	Stores repositories locally	Stores repositories remotely
Collaboration	Can be used to collaborate with others without GitHub	Requires GitHub for collaboration

File States(حالة الملف) in Git

- Untracked(لم يتم تعقبها)
  - Git does not track changes.
- Tracked (تتبع)
  - Git is tracking changes (unmodified, modified, staged)
- Ignored:
  - Git ignores changes (e.g., temporary files).



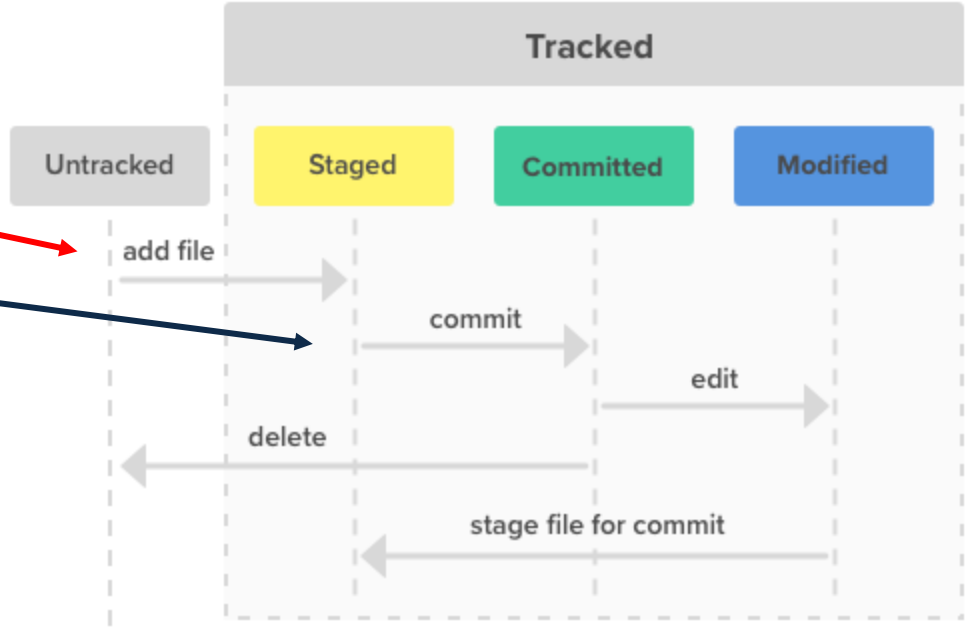
File States(حالة الملف) in Git

- The following Git commands can be used to manage the state of files in your repository:
  - **git add**: Adds files to the staging area. Staged files are the files that will be included in the next commit.
  - **git commit**: Creates a new commit with the staged changes.
  - **git status**: Shows the state of your working directory and staging area.
  - **git checkout**: Checks out a file from the repository to your working directory.
  - **git rm**: Removes a file from the repository.
  - **git mv**: Moves or renames a file in the repository.
  - **git reset**: Resets the staging area or working directory to a previous state.

File State(حالة الملف) in Git

- The following example shows how to use the git add and git commit commands to add a new file to the repository and commit it:

```
# Add the new file to the staging area
git add new_file.txt

# Commit the staged changes
git commit -m "Added new file"
```



- The following example shows how to use the git status command to see the state of your working directory and staging area:

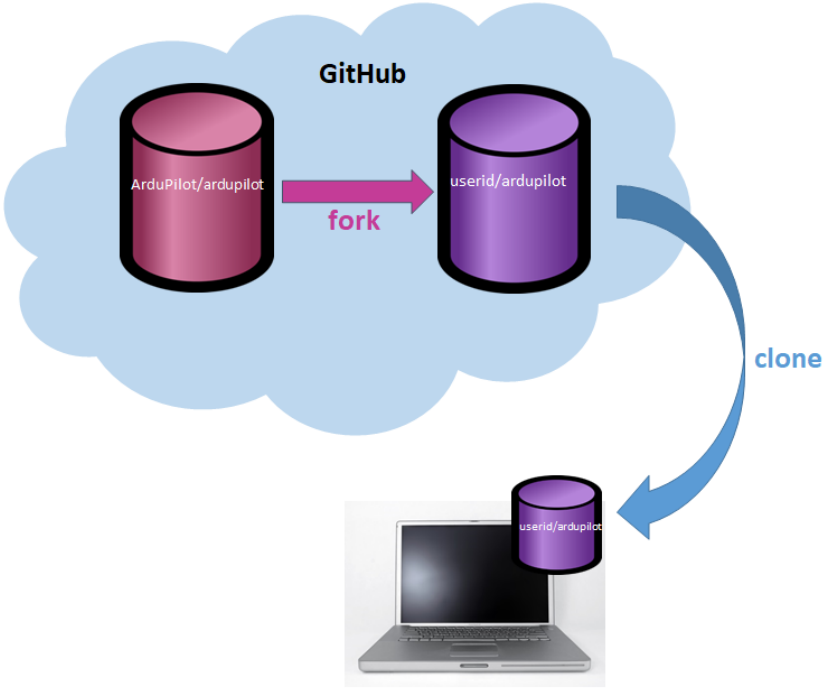
```
git status
```

Cloning(استنساخ) repo in Git

- Cloning a Git repository is the process of creating a local copy of a remote repository.
- The git clone command takes the URL of the remote repository as an argument and creates a local copy of the repository in the current directory.

git clone https://github.com/bard/bard.git

- This would create a local copy of the repository called bard in the current directory.



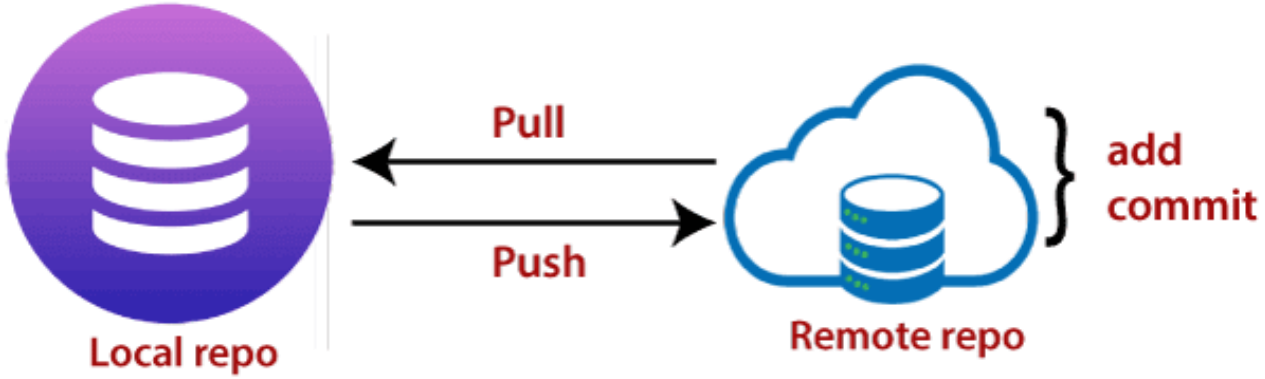
Cloning(استنساخ) repo in Git

- The following example shows how to use the git clone command to clone a repository:

```
# Clone the repository to the current directory
git clone https://github.com/bard/bard.git

# Navigate to the cloned repository
cd bard

# List the contents of the repository
git status
```

Push(دفع) and Pull(سحب) in Git

- Pushing is the process of uploading your local commits to a remote repository.
- Pulling is the process of downloading the latest commits from a remote repository to your local repository and merging them into your local branch.



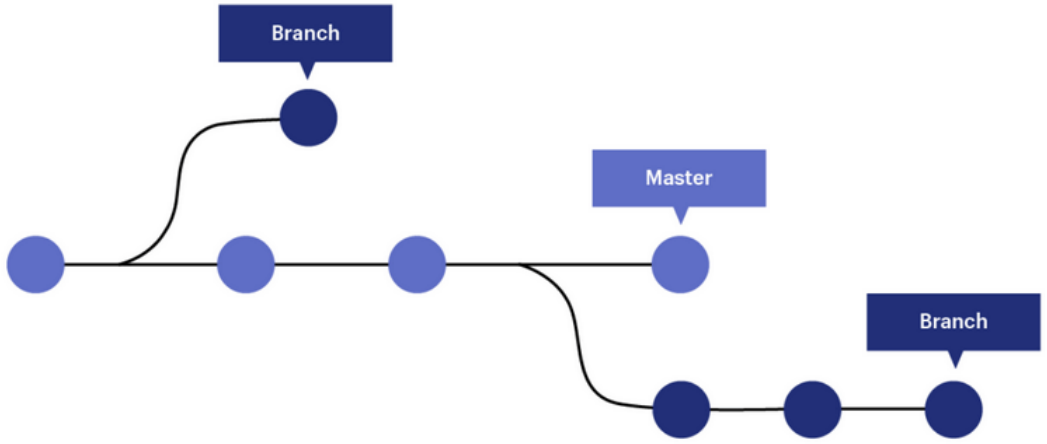
Push(دفع) and Pull(سحب) in Git

- The following example shows how to use the git push and git pull commands to collaborate with another developer:

```
# Developer 1 clones the remote repository to their local computer
git clone https://github.com/developer2/my-repo.git

# Developer 1 makes changes to the code and commits them
git add .
git commit -m "My changes"

# Developer 1 pushes their local commits to the remote repository
git push origin main

# Developer 2 pulls the latest changes from the remote repository
git pull origin main

# Developer 2 sees the changes that Developer 1 made and can continue working on the code
```

Branching(فرع) in Git

- It is a way to isolate changes from the main development line.
- The following Git commands can be used for branching:
  - **git branch**: Creates a new branch.
  - **git checkout**: Switches to a different branch.
  - **git merge**: Merges two branches together.
  - **git rebase**: Applies one branch's commits to another branch.



Branching(فرع) in Git

- The following example shows how to use the git branch, git checkout, and git merge commands to develop a new feature on a branch:

```
# Create a new branch for the new feature
git branch new-feature

# Switch to the new branch
git checkout new-feature

# Make changes to the code and commit them
git add .
git commit -m "Made changes for new feature"

# Push the changes to the remote repository
git push origin new-feature
```

Branching(فرع) in Git

```
# Switch back to the main branch
git checkout main

# Merge the new feature branch into the main branch
git merge new-feature

# Push the merged changes to the remote repository
git push origin main

# Delete the new feature branch
git branch -d new-feature
```

Summary

- **Version Control** tracks changes to files over time.
- **Git** uses local repositories, while **GitHub** hosts remote repositories.
- Files can be in **various states**: untracked, tracked, or ignored.
- **Cloning** copies remote repositories locally.
- **Push and Pull** commands manage data transfer between local and remote repositories.
- **Branching** helps manage different development segments.

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Thank You